INTERNAL MEMO · INFRASTRUCTURE COST REVIEW

AWS Secrets Manager: Cost Estimate & Recommendations

A review of how FieldLoop stores encrypted contractor data, what it costs today, what it will cost as the platform scales, and the safeguard now in place to keep that cost under control.

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SUMMARY

The short version

FieldLoop encrypts every contractor's sensitive field — EIN, SSN, date of birth, home address, bank routing/account numbers, Stripe ID, backup card — as its own entry in AWS Secrets Manager. That gives strong per-field security, but it also means cost scales directly with contractor count: **roughly $3.20 per fully-onboarded contractor, per month**, forever, for as long as that contractor's secrets exist.

At today's scale that's trivial. At 5,000 contractors it's **$16,000 a month** for secret storage alone. Two things now address that: a cleanup of stale/unnecessary secrets (done), and a new *terminate-and-purge* control that automatically deletes a contractor's secrets the moment they're terminated, so cost stops accruing for accounts that are no longer active (built this week — detailed below).

3	$1.20	8	$3.20
secrets in the account today	current monthly Secrets Manager bill	secrets per fully-onboarded contractor	cost added per contractor, per month

BACKGROUND

Why costs looked high before this review

Before this cleanup, the account held **100 secrets** — three core application credentials, and 97 tied to individual contractors. Investigation traced 86 of those to disposable test data generated during earlier QA/testing rounds (accounts named things like "E2E Test Contractor" and "Playwright Full Run"), and a further 11 to four contractor records that weren't clearly test data. Both groups have now been removed. **The account currently holds exactly the three secrets the application itself needs to run** — nothing tied to a contractor remains, which is the correct baseline to project forward from.

COST MODEL

What actually drives the bill

AWS Secrets Manager charges two ways: a flat **$0.40 per secret, per month** it exists (regardless of whether it's ever read), plus **$0.05 per 10,000 API calls** made against it. In FieldLoop's usage pattern — staff occasionally revealing a field, or the app writing one during onboarding — that second charge stays under a few dollars a month even at large scale, so storage is what matters.

Each fully-onboarded contractor can have up to **8 secret-backed fields**: EIN; date of birth, SSN, and home address for the principal; routing number, account number, Stripe account ID, and backup card for financials. Not every contractor fills in every field, but 8 is the realistic ceiling once onboarding is complete — and the number this estimate uses.

`8 secrets × $0.40/secret/month = $3.20 per contractor, per month`, compounding with every additional contractor onboarded and never onboarded off.

PROJECTION

Cost at scale

Projected purely from contractor count, assuming the 8-secret ceiling per contractor:

ACTIVE CONTRACTORS	SECRETS STORED	STORAGE / MONTH	ALL-IN / MONTH
50	400	$160	$161 (+API calls)
200	1,600	$640	$641
1,000	**8,000**	**$3,200**	**$3,202**
5,000	40,000	$16,000	$16,004

API-call costs are estimated from typical reveal/audit activity and stay under $5/month even at 5,000 contractors — storage is the number worth watching.

NOW LIVE

Cost control: automatic purge on termination

Previously, nothing removed a contractor's secrets when that contractor left the platform — cost only ever went up. As of this week, terminating a contractor from the Registration Dashboard (admin-only, in Step 8's new "Danger zone") now:

- Sets the contractor's status to **terminated** — their record and full audit history are kept, nothing about their history disappears.
- Permanently deletes every one of their secrets from Secrets Manager — up to 8 per contractor, stopping the $3.20/month charge for good.
- Is logged to the audit trail with a count of how many secrets were purged, so the action is always traceable.

> WHY KEEP THE RECORDS BUT DELETE THE SECRETS
> A terminated contractor's PII/bank data will never need to be revealed again, so there's no reason to keep paying to store it or keep it exposed — but the contractor row and audit log stay, satisfying recordkeeping and compliance needs that a full data wipe would break.

This doesn't reduce cost for contractors who remain active — it prevents cost from accumulating *indefinitely* for the ones who churn, which over time is the difference between the projection above and a materially larger number.

LOOKING FURTHER OUT

When to reconsider the architecture

Termination cleanup slows the growth curve; it doesn't change its slope. If active-contractor count keeps climbing, the per-field-secret model itself becomes the thing to revisit — and there's a materially cheaper alternative once it's worth the engineering time.

AWS KMS envelope encryption replaces "one Secrets Manager entry per sensitive field" with "one encryption key for the whole application," used to encrypt each field's value directly into a column in the existing database. Pricing is flat: **$1/month per key** plus **$0.03 per 10,000 encrypt/decrypt calls** — cost stops scaling with contractor count almost entirely. The tradeoff is engineering effort: it means new database columns, a migration script, and giving up Secrets Manager's built-in secret versioning UI — a real but bounded, one-time lift.

Now – ~200 contractors	**Stay on Secrets Manager** Cost stays under ~$650/month. The termination-purge control keeps it from creeping higher than active headcount justifies. No architecture change needed yet.
~300–500 contractors	**Begin scoping the KMS migration** Cost is approaching ~$1,000–1,600/month and still climbing. This is the point to size the engineering work, not the point to have already finished it.
1,000+ contractors	**Migrate before this becomes a real line item** At this scale Secrets Manager costs $3,200+/month against a KMS alternative in the low single digits. The gap only widens from here.

> NOT URGENT TODAY
>
> At current scale this is a planning input, not a fire to put out. The recommendation is to revisit this document as contractor count approaches the 300–500 range, rather than migrate preemptively.

RECOMMENDATIONS

In short

- No action needed on current spend — it's $1.20/month today and the cleanup that produced that number is done.
- The termination-purge safeguard is built, deployed, and verified against the live system — future contractor churn will no longer accumulate cost indefinitely.
- Treat 300–500 active contractors as the trigger to scope a KMS envelope-encryption migration, not something to build now.

Figures based on AWS Secrets Manager published pricing, us-east-1, as of this report's date.